



Robust motion control for an underactuated wheeled bipedal robot utilizing sliding mode strategy^{☆,☆☆}

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ABSTRACT

In recent years, wheeled bipedal robots (WBRs) have become one of the frontier fields of robotic research, attracting immense interest from scholars globally. They can not only move rapidly on flat ground through wheels but also possess satisfactory adaptability to uneven terrain due to the introduction of legs. With concise structure and strong agility, WBRs have broad prospects of application in logistics, routing inspection, home service, and so on. However, the inherent underactuation of such robots presents a significant challenge in terms of balance control, particularly in the face of uncertainties and external disturbances. To address this problem, this paper presents a sliding mode control strategy for WBRs, which guarantees robust balance performance even under the influence of various external disturbances. Specifically, the dynamic equations of WBRs are first rearranged in cascaded form, facilitating targeted control design. After that, the wheel torque and leg supporting force are carefully devised to ensure that the sliding surfaces converge to zero within a fixed time. Rigorous Lyapunov-based analysis has shown that the desired equilibrium point is asymptotically stable. Finally, extensive hardware experiments are undertaken, providing convincing evidence of the superior balance control performance achieved by the proposed method.

1. Introduction

Over the past decades, mobile robots have been playing an increasingly important role in the daily life and many industrial fields. Mostly, these robots take the form of wheeled designs, such as sweeping robots, service robots, and even advanced lunar rovers. The popularity of wheeled robots stems from their simplicity, reliability, and the high-efficiency movement they offer, making them ideal for a wide range of tasks. Nevertheless, during the process of application, researchers have noticed certain limitations of the current wheeled robots. For example, they cannot adapt to very complex terrain, which significantly restricts their potential application scenarios. Hence, legged robots are investigated as an alternative (Hamed, Kamidi, Ma, Leonessa, & Ames, 2020; Kuo et al., 2024; Lee, Lee, Lee, & Park, 2022; Takahashi, Shibata, & Nonaka, 2020; Wang & Lin, 2022). Biped humanoid robot such as Atlas (2017), robot dogs such as Cheetah (Seok et al., 2015), AnyMal (Shi et al., 2021), are all typical representatives. By means

of walking, jumping, pacing, etc., they have demonstrated remarkable terrain adaptability, making them highly sought-after for various applications. Recently, significant efforts have been made to deploy these legged robots for various applications, including routing inspection, search and rescue missions, as well as military operations. Unfortunately, these legged robots also present some natural deficiency. To be specific, their movements require to frequently lift and lay down the legs. On the one hand, such mode of motion is of low efficiency, which greatly limits the endurance and range of robots. Besides, the average speed of legged robots is relatively slow, which fails to meet demand under certain circumstances. On the other hand, the pursuit of swift leg motion often comes at the cost of reduced leg strength, due to the finite motor power available. This trade-off limits the carrying capacity of some legged robots, as they may struggle to support heavy loads.

In view of the above problems, many researchers have embarked on an innovative path by combining the strengths of both designs, resulting in the emergence of the wheeled bipedal robot (WBR). Recently,

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^{☆☆} Video of the tests for the proposed method is available at <https://youtu.be/K6T56vM37jA>.

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WBR has become one of the frontier fields of robotic research, and has attracted world-wide attention. The WBR's unique design allows it to seamlessly transition between two modes of locomotion: it can swiftly traverse flat surfaces using its wheels, capitalizing on their efficiency and speed. Moreover, by incorporating legs, the WBR gains remarkable adaptability to uneven and complex terrains, overcoming the limitations of traditional wheeled robots. With concise structure and strong agility, WBR has broad prospects of application in logistics, routing inspection, home service, and so on. To this end, an important issue is the balance control of WBR, which is very challenging due to the system underactuation nature (Lu & Fang, 0000; Miao, Yang, & Ouyang, 2023; Sandoval, Moyrón, Kelly, Santibáñez, & Moreno-Valenzuela, 2021; Xian & Yang, 2021; Yu et al., 2023). Till now, this topic has been extensively studied, with many remarkable results published (Cao, Lu, Liu, Liu, & Guo, 2022; Dinev, Xin, Merkt, Ivan, & Vijayakumar, 2020; Li, Zhou, Feng, Zhang, & Fu, 2018, 2019; Lu, Cao, Hao, Lin, & Fang, 2022; Lu & Fang, 2021). Specifically, Handle from Boston Dynamics is one of the most successful WBRs (Boston Dynamics, 2017). It has shown very powerful balancing ability not only in different terrains, but also in different tasks (such as jumping, cargo delivering, obstacle avoiding). However, unfortunately, the control strategy of Handle remains largely unknown to date. Different from Handle which has very sophisticated electromechanical design, Ascento proposed by Eidgenössische Technische Hochschule Zürich (ETH) has relatively concise structure. It consists of only four motors (two hip motors and two hub motors), and weighs about 10 kg. By utilizing an efficient WBC (whole body control) framework, Ascento manages to move agilely on the ground, jump, and even recover from fall condition (Klemm et al., 2020, 2019). The underlying balancing control algorithm of Ascento is the LQR (Linear Quadratic Regulator) method. To further improve the obstacle crossing ability, ETH has recently launched a new version of robot - Ascento Pro. Equipped with more powerful motors and bigger wheels, Ascento Pro successfully balances itself even when climbing the stairs. Southern University of Science and Technology (Shenzhen, China) has designed a WBR named "Nezha". In Liu, Chen, Wensing, and Zhang (2021), an ICI (instantaneous capture input) - based controller with online optimal feedback gains is proposed for the balancing control of Nezha, which can adapt to variable-height of the robot. Tencent Technology Co., LTD (Shenzhen, China) has developed an Ollie robot, which can accomplish difficult movements such as backflip (Wang et al., 2021). In Xu, Guo, and Lee (2014), an integral sliding mode controller is designed for an underactuated two-wheeled mobile robot. The controller has good robustness toward both matched and unmatched disturbances. However, the robot leg length is considered as constant in this paper, which limits its application. By solving a class of boundary value problems with internal constraints, an online trajectory planning strategy is proposed for WBR in Kim and Kwon (2020), which can improve the dynamic agility of robots. With ability to solve variable constraints, model predictive control (MPC) has been extensively applied to mobile robot platforms in recent years (Chen, Li, Kong, & Ke, 2018; Grandia, Farshidian, Ranftl, & Hutter, 2019; Liu et al., 2022; Xiao et al., 2016). In Xin and Vijayakumar (2020), an online dynamic hybrid control strategy based on MPC is proposed, which enables the robot to travel through chaotic environment. By using the decoupling unit, two independent MPC algorithms are designed in Minouchehr and Hosseini (2015) to control the two-wheeled inverted pendulum (TWIP) robot. However, the corresponding control performance is seriously affected by modeling error, time-varying interference and other factors. To further improve the robustness against disturbances, an adaptive MPC method based on linear time-varying Kalman filter is designed in Zad and Ulasz (2017). There are also many interesting researches studying the jumping movements of WBR (Chen et al., 2021; Hao, Lu, Cao, et al., 2022), which present new challenges in controller design.

Summarizing the above facts, the challenge of balancing control for WBR, particularly when confronted with variable leg lengths, remains an unsolved problem. Existing results often simplify the WBR

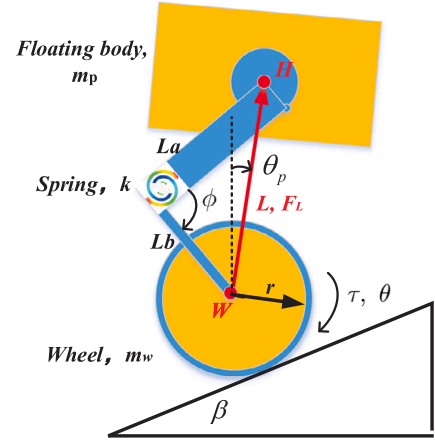


Fig. 1. Schematic illustration of the WBR system.

as an inverted pendulum, focusing solely on planar motion, while neglecting critical factors such as system uncertainties and external disturbances. This simplification can lead to suboptimal performance in real-world applications. To deal with these issues, a nonlinear sliding mode controller is proposed for WBR in this paper, which guarantees robust balance performance in motion even under the influence of various external disturbances. Specifically, with appropriate state transformations, the dynamic equations of the WBR are first restructured into a cascaded form, facilitating subsequent control design. After that, the wheel torque and leg supporting force are elaborately designed, which drive the sliding surfaces to zero in fixed time. Rigorous Lyapunov-based analysis has shown that, the desired equilibrium point is asymptotically stable. Finally, sufficient hardware experiments are performed to validate the balance control performance of the proposed method.

The contributions of this paper are summarized as follows:

- It elaborately considers different ground slopes, time-varying leg length, and other factors of WBR in practical working conditions. Besides, fewer simplifications are adopted in stability analysis compared with existing works, which guarantees more reliable performance in real applications.
- The proposed method is instructive for tackling similar problems, particularly in that it can be readily adapted and extended to facilitate 3-dimensional motion control of WBRs, significantly alleviating the challenges associated with controller design and analysis. Experiment results are provided to demonstrate that the robot can move agilely and smoothly on different terrains.
- The proposed method achieves robust balancing performance. Particularly, it is not only insensitive to parameter uncertainties, but also able to quickly eliminate the influence of external disturbances. Besides, the chattering problem is greatly alleviated with the help of sliding mode observers and the virtual input.

The rest of this paper is organized as follows. In Section 2, the WBR platform and dynamic equations are introduced. Sections 3–4 respectively detail the controller design process and the corresponding stability analysis. In Section 5, hardware experiment results are provided to demonstrate the performance of the proposed method. Finally, some conclusions are presented in Section 6.

2. Problem statement

Fig. 1 is the schematic illustration of a WBR. As can be observed, the robot is traversing an uneven terrain, with the slope angle denoted by β . m_w represents the mass of the wheel, whose radius is denoted

by r . m_p is the floating body mass. The upper leg L_a and the lower leg L_b connect the wheel and robot body. On the knee joint, a spring with the coefficient k is mounted to help the robot gain better load-bearing capacity. L stands for the distance between the center of wheel (point W) and the center of hip joint (point H). One can use L to roughly represent the robot leg length. As for F_L , it is the generalized supporting force along the leg (the direction of F_L is shown with a red arrow in Fig. 1). This force is an equivalent force, which is generated by the knee joint motor. In particular, the robot leg length L can be conveniently adjusted by the knee motor. Meantime, the angle ϕ between L_a and L_b will also change. The angular displacement of the wheel is denoted by $\theta(t)$, and $\theta_p(t)$ stands for the incline angle of the robot. $\tau(t)$ is the wheel motor torque.

The dynamic equations of WBR are represented as¹

$$M(q)\ddot{q} + V(q, \dot{q}) + G(q) = S^T(u + d), \quad (1)$$

where $q = [\theta \ L \ \theta_p]$ is the system state, $M(q)$, $V(q, \dot{q}) \in \mathbb{R}^{3 \times 3}$, $G(q) \in \mathbb{R}^3$ denote the inertia matrix, the centripetal-Coriolis matrix, and the gravity vector. $u \in \mathbb{R}^2$ represents the control input vector. $d \in \mathbb{R}^2$ stands for the disturbance vector. The explicit expressions of the above matrices and vectors are given as follows:

$$\begin{aligned} M(q) &= \begin{bmatrix} m_w r^2 + m_p r^2 + J_w & m_p r \sin(\beta + \theta_p) & m_{13} \\ m_p r \sin(\beta + \theta_p) & m_p & 0 \\ m_{31} & 0 & m_p L^2 \end{bmatrix}, \\ V(q, \dot{q}) &= \begin{bmatrix} 0 & m_p r \cos(\beta + \theta_p) \dot{\theta}_p & v_{13} \\ 0 & 0 & -m_p L \dot{\theta}_p \\ 0 & m_p L \dot{\theta}_p & m_p L \dot{L} \end{bmatrix}, \\ G(q) &= \begin{bmatrix} (m_p + m_w) g r \sin \beta \\ m_p g \cos \theta_p - k L \Delta L \\ -m_p g L \sin \theta_p \end{bmatrix}, \quad u = \begin{bmatrix} \tau \\ F_L \end{bmatrix}, \\ d &= \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}, \end{aligned} \quad (2)$$

where

$$\begin{aligned} m_{31} &= m_{13} = m_p r L \cos(\beta + \theta_p), \\ v_{13} &= m_p r \cos(\beta + \theta_p) \dot{L} - m_p r L \sin(\beta + \theta_p) \dot{\theta}_p, \\ \Delta L &= \frac{\phi_0 - \phi}{L_a L_b \sin \phi}. \end{aligned}$$

For convenience of description, (1) is expanded as

$$\begin{aligned} (m_w r^2 + m_p r^2 + J_w) \ddot{\theta} + m_p r \sin(\beta + \theta_p) \ddot{L} \\ + m_p r L \cos(\beta + \theta_p) \ddot{\theta}_p + 2 m_p r \cos(\beta + \theta_p) \dot{L} \dot{\theta}_p \\ - m_p r L \sin(\beta + \theta_p) \dot{\theta}_p^2 + (m_p + m) g r \sin \beta = \tau + d_1, \end{aligned} \quad (3)$$

$$\begin{aligned} m_p r \sin(\beta + \theta_p) \ddot{\theta} + m_p \ddot{L} - m_p L \dot{\theta}_p^2 + m_p g \cos \theta_p \\ - k L \Delta L = F_L + d_2, \end{aligned} \quad (4)$$

$$\begin{aligned} m_p r L \cos(\beta + \theta_p) \ddot{\theta} + m_p L^2 \ddot{\theta}_p + 2 m_p L \dot{L} \dot{\theta}_p \\ - m_p g L \sin \theta_p = -\tau. \end{aligned} \quad (5)$$

Adding (3) and (5) gives rise to

$$\begin{aligned} [m_w r^2 + m_p r^2 + J_w + m_p r L \cos(\beta + \theta_p)] \ddot{\theta} \\ + m_p r \sin(\beta + \theta_p) \ddot{L} + [m_p r L \cos(\beta + \theta_p) + m_p L^2] \ddot{\theta}_p \\ + 2[m_p \cos(\beta + \theta_p) + m_p L] \dot{L} \dot{\theta}_p - m_p \sin(\beta + \theta_p) L \dot{\theta}_p^2 \\ + (m_p + m) g \sin \beta - m_p g L \sin \theta_p = d_1. \end{aligned} \quad (6)$$

¹ WBR is a kind of underactuated system, which means it has less independent control inputs than the degrees of freedom (DOFs). In this case, the two DOFs θ and θ_p share the same control input τ . Specifically speaking, the stator of wheel hub motor is mounted on the robot body (lower leg), and its rotor is mounted on the wheel. Hence, when the motor outputs torque τ to drive the wheel forward, a reaction torque $-\tau$ is generated to drive the robot body to incline backward.

To construct appropriate sliding surfaces, the couplings between system states must be carefully analyzed and utilized. However, the original system dynamics (3)–(5), (6) are very complicated, which brings much difficulty in achieving the above goal. Therefore, one decides to make some model transformations by defining new state variables. After very elaborate analysis, χ_1 to χ_4 are introduced as follows:

$$\begin{aligned} \chi_1 &= \lambda(\theta - \theta_d) + m_p r L [\sin(\beta + \theta_p) - \sin(\beta + \theta_{pd})] \\ &\quad + m_p L^2 (\theta_p - \theta_{pd}), \\ \chi_2 &= \lambda \dot{\theta} + m_p r [\dot{L} \sin(\beta + \theta_p) + L \cos(\beta + \theta_p) \dot{\theta}_p] \\ &\quad + m_p L^2 \dot{\theta}_p, \\ \chi_3 &= m_p r [L \sin(\beta + \theta_p) - L_d \sin(\beta + \theta_{pd})], \\ \chi_4 &= m_p r [\dot{L} \sin(\beta + \theta_p) + L \cos(\beta + \theta_p) \dot{\theta}_p] + m_p L^2 \dot{\theta}_p. \end{aligned} \quad (7)$$

In (7), θ_d , L_d , θ_{pd} respectively denote the desired values of θ , L , θ_p . On this basis, one can rearrange (6) as the following cascaded form:

$$\begin{aligned} \dot{\chi}_1 &= \chi_2 + \xi_1, \\ \dot{\chi}_2 &= \chi_3 + \xi_2, \\ \dot{\chi}_3 &= \chi_4 + \xi_3, \\ \dot{\chi}_4 &= -G - \lambda \ddot{\theta} + d_1, \end{aligned} \quad (8)$$

where the explicit expressions of λ , G , ξ_i , $i = 1, 2, 3$ are provided as follows:

$$\begin{aligned} \lambda &= m_w r^2 + m_p r^2 + J_w + m_p r L \cos(\beta + \theta_p), \\ G &= (m_p + m) g \sin \beta - m_p g L \sin \theta_p, \\ \xi_1 &= \dot{\lambda}(\theta - \theta_d) + m_p r \dot{L} [\sin(\beta + \theta_p) - \sin(\beta + \theta_{pd})] \\ &\quad + 2 m_p L \dot{L} (\theta_p - \theta_{pd}), \\ \xi_2 &= \dot{\lambda} \dot{\theta} - G + m_p g [L \sin(\beta + \theta_p) - L_d \sin(\beta + \theta_{pd})] + d_1, \\ \xi_3 &= -m_p L^2 \dot{\theta}_p. \end{aligned} \quad (9)$$

The following assumptions are adopted before proceeding with the subsequent analysis:

Assumption 1. The external disturbances are all bounded in the sense of

$$|d_i| \leq B_i, \quad i = 1, 2, \quad \forall t \in \mathbb{R}^+, \quad (10)$$

where B_i are upper bounds.

Assumption 2. Both ξ and χ are complex nonlinear functions, hence, it is extremely difficult to get analytical solution for the relationship between them. On the other hand, both ξ and χ are functions of q , $\dot{q}$, one can reasonably deduce/assume that the following relationship holds for them (Lu, Fang, & Sun, 2018):

$$\|\xi\| \leq \alpha_1 \|\chi\| + \alpha_2 \|\chi\|^2, \quad (11)$$

where $\xi = [\xi_1 \ \xi_2 \ \xi_3]^T$, $\chi = [\chi_1 \ \chi_2 \ \chi_3]^T$, α_1 , α_2 are positive constants.

Remark 1. To facilitate controller design, one wants to first rearrange (3)–(5) into cascaded form, which is actually very difficult for such a nonlinear underactuated system. To achieve this goal, it necessitates a thorough understanding of the system's dynamic characteristics, often requiring extensive analysis and numerous iterations due to the inherent difficulties. Drawing inspiration from the form of (6), one notices that the time derivative of χ_2 can be represented by $m_p g L \sin \theta_p$ (the last term on the left of Eq. (6)), and further by χ_3 . Since χ_3 does not contain velocity signals, differentiating it with respect to time will not introduce second order derivatives, which implies that χ_4 (component of $\dot{\chi}_3$) can be further differentiated for controller design. This is exactly what one requires to construct a cascaded system such as (8). Once χ_2 and χ_3 are appropriately defined and determined, χ_1 and χ_4 can be systematically obtained through integration or differentiation, depending on the context and requirements of the controller design.

3. Sliding mode controller design

In this section, one first defines a virtual control input, which is generated by a novel second order sliding mode strategy. On this basis, the real control inputs τ and F_L are further constructed. To enhance the robustness and performance of the control system, one leverages sliding mode observers to estimate and compensate for disturbances that may arise during the operation of the WBR. These observers not only provide accurate estimates of the disturbances but also play a pivotal role in alleviating the chattering problem, which is a common issue in traditional sliding mode control (SMC) approaches.

3.1. Design of the virtual control input v

First, one defines the following variable:

$$x_1 = \rho_1 \chi_1 + \rho_2 \chi_2 + \rho_3 \chi_3 + \chi_4, \quad (12)$$

where ρ_1, ρ_2, ρ_3 are positive control parameters. Taking the time derivative of x_1 gives rise to

$$\begin{aligned} \dot{x}_1 &= \rho_1 \dot{\chi}_1 + \rho_2 \dot{\chi}_2 + \rho_3 \dot{\chi}_3 + \dot{\chi}_4 \\ &= \rho_1(\chi_2 + \xi_1) + \rho_2(\chi_3 + \xi_2) \\ &\quad + \rho_3(\chi_4 + \xi_3) - G + v + d_1, \end{aligned} \quad (13)$$

where $v = -\lambda \ddot{\theta}$ is considered as the virtual control input. Subsequently, one would like to design an appropriate virtual control input v , so as to balance the WBR system. However, traditional SMC design concept may cause the chattering problem, and it is difficult to guarantee finite time convergence. To avoid the above drawbacks, a more sophisticated second order sliding mode strategy is presented. To this end, one further designs the following variable:

$$x_2 = f(\chi, \chi_4, \xi) - G + v, \quad (14)$$

where $f(\chi, \chi_4, \xi) = \rho_1(\chi_2 + \xi_1) + \rho_2(\chi_3 + \xi_2) + \rho_3(\chi_4 + \xi_3)$. Differentiating x_2 with respect to time, one obtains

$$\dot{x}_2 = \frac{\partial f}{\partial \chi} \dot{\chi} + \frac{\partial f}{\partial \chi_4} (d_1 + v - G) + \frac{\partial f}{\partial \xi} \frac{\partial \xi}{\partial \chi} \dot{\chi} - \dot{G} + \dot{v}. \quad (15)$$

To facilitate description, one adopts the following notations:

$$h_1 = \frac{\partial f}{\partial \chi} \dot{\chi} - \frac{\partial f}{\partial \chi_4} G + \frac{\partial f}{\partial \xi} \frac{\partial \xi}{\partial \chi} \dot{\chi} - \dot{G}, \quad \frac{\partial f}{\partial \chi_4} = f_4. \quad (16)$$

In (16), h_1 and f_4 are both known functions. Therefore, one obtains the following results from (13)–(16):

$$\begin{aligned} \dot{x}_1 &= x_2 + d_1, \\ \dot{x}_2 &= h_1 + f_4(v + d_1) + \dot{v}. \end{aligned} \quad (17)$$

Regarding (17), the following sliding surface is designed:

$$s_1 = (x_2 + \hat{d}_1)^{q_1/p_1} + a_1(x_2 + \hat{d}_1) + c_1 x_1, \quad (18)$$

where a_1, c_1 are positive control gains to be determined, $p_1, q_1 > 0$ are odd integers satisfying $1 < q_1/p_1 < 2$. In (18), $\hat{d}_1$ is the estimate of d_1 , which is generated by the following sliding mode observer:

$$\begin{aligned} \dot{\hat{x}}_1 &= x_2 + \omega_0, \quad \omega_0 = -\kappa_0 B_1^{\frac{1}{2}} |\hat{x}_1 - x_1|^{\frac{1}{2}} \text{sgn}(\hat{x}_1 - x_1) + \hat{d}_1, \\ \dot{\hat{d}}_1 &= \omega_1, \quad \omega_1 = -\kappa_1 B_1 \text{sgn}(\hat{d}_1 - \omega_0), \end{aligned} \quad (19)$$

where $\kappa_0, \kappa_1 > 0$ are positive control parameters. According to Yang, Li, Su, and Yu (2013), one derives that $\hat{d}_1$ converges to d_1 in finite time. Based on the estimation results and the structure of s_1 , the virtual control input v is elaborately designed in the following manner:

$$\begin{aligned} \dot{v} &= -\omega_1 - h_1 - f_4 v - f_4 \hat{d}_1 - \frac{c_1}{\rho_1} (x_2 + \hat{d}_1) \\ &\quad - \frac{1}{2\phi_1} (k_{s_1} |s_1|^{\epsilon_1} \text{sgn}(s_1) + k_{a_1} s_1^{p_1/q_1} + k_{\beta_1} s_1^{2-p_1/q_1}), \end{aligned} \quad (20)$$

where $0 < \epsilon_1 < 1$ is a control parameter, $\phi_1 \geq 0$ denotes $a_1 + \frac{q_1}{p_1} (x_2 + \hat{d}_1)^{\frac{q_1-p_1}{p_1}}$ for conciseness, ω_1 is the estimation of $\dot{d}_1$ shown in (19).

3.2. Design of the real control inputs τ and F_L

From (20), one obtains a continuous virtual control input v , which further gives rise to

$$\ddot{\theta} = -v/\lambda. \quad (21)$$

It is then inferred from (4) that

$$m_p \ddot{L} = F_L + h_2 + d_2, \quad (22)$$

where

$$\begin{aligned} h_2 &= m_p r \sin(\beta + \theta_p) v / \lambda + m_p L \dot{\theta}_p^2 \\ &\quad - m_p g \cos \theta_p + k L \Delta L. \end{aligned}$$

Regarding (22), one defines the following auxiliary variables:

$$x_{l_1} = m_p (L - L_d), \quad x_{l_2} = m_p \dot{L}. \quad (23)$$

Consequently, the following cascaded system is derived:

$$\begin{aligned} \dot{x}_{l_1} &= x_{l_2}, \\ \dot{x}_{l_2} &= F_L + h_2 + d_2. \end{aligned} \quad (24)$$

Similar to (19), an observer is designed to estimate d_2 :

$$\begin{aligned} \dot{\hat{x}}_{l_2} &= F_L + h_2 + \omega_{l_0}, \\ \omega_{l_0} &= -\kappa_{l_0} B_2^{\frac{1}{2}} |\hat{x}_{l_2} - x_{l_2}|^{\frac{1}{2}} \text{sgn}(\hat{x}_{l_2} - x_{l_2}) + \hat{d}_2, \\ \dot{\hat{d}}_2 &= \omega_{l_1}, \quad \omega_{l_1} = -\kappa_{l_1} B_2 \text{sgn}(\hat{d}_2 - \omega_{l_0}), \end{aligned} \quad (25)$$

where $\kappa_{l_0}, \kappa_{l_1} > 0$ are control gains. Based on the aforementioned considerations, the following sliding surface is constructed:

$$s_2 = x_{l_2}^{q_2/p_2} + a_2 x_{l_2} + c_2 x_{l_1}. \quad (26)$$

Accordingly, F_L is designed in the following manner:

$$\begin{aligned} F_L &= -h_2 - \hat{d}_2 - \frac{c_2}{\phi_2} x_{l_2} \\ &\quad - \frac{1}{2\phi_2} (k_{s_2} |s_2|^{\epsilon_2} \text{sgn}(s_2) + k_{a_2} s_2^{p_2/q_2} + k_{\beta_2} s_2^{2-p_2/q_2}), \end{aligned} \quad (27)$$

where $0 < \epsilon_2 < 1$ is a control parameter, $\phi_2 = a_2 + \frac{q_2}{p_2} x_{l_2}^{(q_2-p_2)/p_2} > 0$.

Substituting the estimation results of (25) into (22), one derives

$$\ddot{L} = (F_L + h_2 + \hat{d}_2)/m_p. \quad (28)$$

Since $v = -\lambda \ddot{\theta}$ and F_L can be conveniently derived from (20), (27), $\ddot{L}$ and $\ddot{\theta}$ are also known functions (see (21), (28)). To further obtain $\ddot{\theta}_p$, one rearranges (6) as follows:

$$\lambda \ddot{\theta} + \lambda_L \ddot{L} + \lambda_p \ddot{\theta}_p + \Phi = d_1, \quad (29)$$

where

$$\begin{aligned} \lambda_L &= m_p r \sin(\beta + \theta_p), \\ \lambda_p &= m_p r L \cos(\beta + \theta_p) + m_p L^2, \\ \Phi &= 2[m_p \cos(\beta + \theta_p) + m_p L] \dot{L} \dot{\theta}_p \\ &\quad - m_p \sin(\beta + \theta_p) L \dot{\theta}_p^2 + G. \end{aligned} \quad (30)$$

Using the results from Eqs. (29) and (30), one can derive the expression for $\ddot{\theta}_p$ as follows:

$$\ddot{\theta}_p = \frac{\hat{d}_1 + v - \lambda_L \ddot{L} - \Phi}{\lambda_p}. \quad (31)$$

Summarizing the results of (21), (28), (31), $\ddot{\theta}$, $\ddot{L}$, $\ddot{\theta}_p$ are all known functions. Hence, one can finally obtain the wheel hub motor torque τ as follows:

$$\begin{aligned} \tau &= m_p r L \cos(\beta + \theta_p) v / \lambda - 2m_p L \dot{L} \dot{\theta}_p + m_p g L \sin \theta_p \\ &\quad - \frac{m_p L^2}{\lambda_p} \left[\hat{d}_1 + v - \frac{\lambda_L (F_L + h_2 + \hat{d}_2)}{m_p} - \Phi \right]. \end{aligned} \quad (32)$$

4. Stability analysis

In this section, rigorous Lyapunov-based analysis is performed to show that, the proposed method guarantees fixed-time convergence of the sliding surfaces. After that, the desired equilibrium point is proved to be asymptotically stable.

4.1. Convergence of s_1

First, one designs the following non-negative Lyapunov function candidate:

$$V_1(t) = s_1^2 \geq 0. \quad (33)$$

Differentiating (33) with respect to time, the following results are obtained:

$$\dot{V}_1(t) = 2s_1[c_1(x_2 + d_1) + \rho_1(\omega_1 + h_1 + f_4v + f_4d_1 + \dot{v})], \quad (34)$$

where (17)–(19) are utilized. Substituting the virtual input (20) into (34), one derives

$$\begin{aligned} \dot{V}_1(t) = 2s_1 & \left[(c_1 + \rho_1 f_4)e_{d_1} - \frac{1}{2}k_{s_1}|s_1|^{\epsilon_1} \text{sgn}(s_1) \right. \\ & \left. - \frac{1}{2}k_{\alpha_1}s_1^{p_1/q_1} - \frac{1}{2}k_{\beta_1}s_1^{\frac{2q_1-p_1}{q_1}} \right] \\ & = -k_{s_1}|s_1|^{1+\epsilon_1} - k_{\alpha_1}s_1^{\frac{p_1+q_1}{q_1}} - k_{\beta_1}s_1^{\frac{3q_1-p_1}{q_1}} \\ & \quad + 2(c_1 + \rho_1 f_4)e_{d_1}s_1, \end{aligned} \quad (35)$$

where $e_{d_1} = d_1 - \hat{d}_1$ denotes the estimation error. Since $\hat{d}_1$ converges to d_1 (see Yang et al. (2013)), one can further reduce (35) as follows:

$$\begin{aligned} \dot{V}_1(t) & = -k_{s_1}|s_1|^{1+\epsilon_1} - k_{\alpha_1}s_1^{\frac{p_1+q_1}{q_1}} - k_{\beta_1}s_1^{\frac{3q_1-p_1}{q_1}} \\ & \leq -k_{\alpha_1}s_1^{\frac{p_1+q_1}{q_1}} - k_{\beta_1}s_1^{\frac{3q_1-p_1}{q_1}} \\ & = -(k_{\alpha_1} + k_{\beta_1}\Lambda^2)V_1^{\frac{p_1+q_1}{2q_1}} \\ & \leq 0, \end{aligned} \quad (36)$$

where $\Lambda = V_1^{\frac{q_1-p_1}{2q_1}}$. With some further arrangements, (36) is rewritten as follows:

$$\begin{aligned} \frac{dV_1}{dt} & \leq -(k_{\alpha_1} + k_{\beta_1}\Lambda^2)V_1^{\frac{p_1+q_1}{2q_1}} \\ \Rightarrow \frac{1}{V_1^{\frac{p_1+q_1}{2q_1}}} dV_1 & \leq -(k_{\alpha_1} + k_{\beta_1}\Lambda^2)dt \\ \Rightarrow \frac{2q_1}{q_1 - p_1} d\Lambda & \leq -(k_{\alpha_1} + k_{\beta_1}\Lambda^2)dt \\ \Rightarrow \frac{d\Lambda}{k_{\alpha_1} + k_{\beta_1}\Lambda^2} & \leq -\frac{q_1 - p_1}{2q_1} dt. \end{aligned} \quad (37)$$

Integrating both sides of (37), it is concluded that

$$F(t) - F(0) \leq -\frac{q_1 - p_1}{2q_1} t \Rightarrow F(t) \leq F(0) - \frac{q_1 - p_1}{2q_1} t,$$

where $F(t) = \frac{1}{\sqrt{k_{\alpha_1}k_{\beta_1}}} \arctan(\frac{k_{\beta_1}}{k_{\alpha_1}}\Lambda)$. According to the statements below (18), one derives $\frac{q_1-p_1}{2q_1} > 0$, which implies that $F(t)$ continuously decreases. Since $F < \frac{\pi}{2\sqrt{k_{\alpha_1}k_{\beta_1}}}$ always holds, the convergence time for $F(t)$, which is also the convergence time for Λ and s , is obtained as

$$T_1 \leq \frac{q_1\pi}{(q_1 - p_1)\sqrt{k_{\alpha_1}k_{\beta_1}}}. \quad (38)$$

It should be noticed that, the convergence time T_1 is not related to initial system state.

4.2. Convergence of s_2

Choose the following Lyapunov function candidate:

$$V_2(t) = s_2^2 \geq 0. \quad (39)$$

Differentiating (39) with respect to time gives rise to

$$\begin{aligned} \dot{V}_2(t) & = 2s_2[c_2x_{l_2} + \frac{q_2}{p_2}x_2^{\frac{q_2-p_2}{p_2}}(F_L + h_2 + d_2)] \\ & = 2s_2[c_2x_{l_2} + \rho_2(F_L + h_2 + d_2)], \end{aligned} \quad (40)$$

where (24), (26) are utilized. By substituting Eq. (27) into (40), one obtains

$$\begin{aligned} \dot{V}_2(t) & = -k_{s_2}|s_2|^{1+\epsilon_2} - k_{\alpha_2}s_2^{\frac{p_2+q_2}{q_2}} - k_{\beta_2}s_2^{\frac{2q_2-p_2}{q_2}} \\ & \quad + 2\rho_2e_{d_2}s, \end{aligned} \quad (41)$$

where e_{d_2} denotes the estimation error. Similar to the analysis of (35)–(38), one derives that s_2 converges to 0 in fixed time.

4.3. Convergence of the system state variables

Since s_1 converges to zero in finite time T_1 , namely,

$$s_1 = (x_2 + \hat{d}_1)^{\frac{q_1}{p_1}} + a_1(x_2 + \hat{d}_1) + c_1x_1 = 0, \quad \forall t > T_1. \quad (42)$$

One can further deduce from (42) that x_1 also converges to 0 in finite time (see Mohammadi Asl, Shabbouei Hagh, Palm, & Handroos, 2019; Razmjooei, Palli, Abdi, Strano, & Terzo, 2022; Razmjooei, Shafiei, Palli, & Ibeas, 2020; Van, 2018), which, along with (12), implies

$$x_1 = 0 \Rightarrow \chi_4 = -(\rho_1\chi_1 + \rho_2\chi_2 + \rho_3\chi_3). \quad (43)$$

Substitute (43) into (8), one derives

$$\begin{aligned} \dot{\chi}_1 & = \chi_2 + \xi_1, \\ \dot{\chi}_2 & = \chi_3 + \xi_2, \\ \dot{\chi}_3 & = -(\rho_1\chi_1 + \rho_2\chi_2 + \rho_3\chi_3) + \xi_3. \end{aligned} \quad (44)$$

To facilitate description, one defines

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\chi_1 & -\chi_2 & -\chi_3 \end{bmatrix}.$$

Hence, (44) could be expressed in the following matrix form:

$$\dot{\chi} = A\chi + \xi. \quad (45)$$

According to (45), one chooses the control parameters ρ_i in the following way:

$$\rho_1 = k^3, \quad \rho_2 = 3k^2, \quad \rho_3 = 3k, \quad (46)$$

where $k > 0$ is a to-be-determined control gain. Therefore, the triple eigenvalue of A can be calculated as $-k$. This configuration is critical because it ensures that all eigenvalues of A have negative real parts, thereby satisfying the Hurwitz stability criterion. On the other hand, to facilitate analysis, one defines the following state transformation:

$$\eta = \Gamma^{-1}\chi, \quad (47)$$

where Γ is an invertible matrix satisfying the following property:

$$\Gamma^{-1}A\Gamma = -J, \quad (48)$$

where J denotes

$$J = \begin{bmatrix} k & -1 & 0 \\ 0 & k & -1 \\ 0 & 0 & k \end{bmatrix}.$$

It could be theoretically proved by the *Theory of Matrices* that, there always exists such matrix as Γ for (48) (see Lu et al. (2018)).

Based on (45)–(48), one further defines the following Lyapunov function candidate:

$$V_3(t) = \eta^T \eta \geq 0.$$

Taking the time derivative of V_3 yields

$$\dot{V}_3(t) = -\eta^T(J^T + J)\eta + 2\eta^T\Gamma^{-1}\xi, \quad (49)$$

wherein $(J^T + J)$ is a positive definite matrix, whose eigenvalues could be calculated as $2k$, $2k \pm \sqrt{2}$. If one chooses $k > \sqrt{2}/2$, then the smallest eigenvalue of $(J^T + J)$ would be $2k - \sqrt{2} > 0$. Therefore, (49) could be further reduced as follows:

$$\begin{aligned} \dot{V}_3(t) &\leq -(2k - \sqrt{2}) \cdot \|\eta\|^2 \\ &\quad + 2\|\eta\| \cdot \|\Gamma^{-1}\| \cdot (\alpha_1\|\chi\| + \alpha_2\|\chi\|^2) \\ &\leq -\gamma_1 \cdot \|\eta\|^2 + \gamma_2 \cdot \|\eta\|^3, \end{aligned} \quad (50)$$

wherein

$$\begin{aligned} \gamma_1 &= 2k - \sqrt{2} - 2\alpha_1 \cdot \|\Gamma^{-1}\| \cdot \|\Gamma\|, \\ \gamma_2 &= 2\alpha_2 \cdot \|\Gamma^{-1}\| \cdot \|\Gamma\|^2. \end{aligned} \quad (51)$$

Based on (51), one elaborately selects a sufficiently large k to ensure that γ_1 remains positive. Besides, $\|\eta(0)\| < \frac{\gamma_1}{\gamma_2}$ should be guaranteed.

Hence, it is concluded that, the initial value of $\dot{V}_3(t)$ is negative, namely, $\dot{V}_3(0) = -\|\eta(0)\|^2 \cdot [\gamma_1 - \gamma_2 \cdot \|\eta(0)\|] \leq 0$. Consequently, $V_3 = \|\eta\|^2$ continuously decreases, which further leads to the fact $\dot{V}_3 < 0$. In this way, $\|\eta\|$ will converge to zero quasi-exponentially, which further implies that $\|\chi\|$ converge to zero quasi-exponentially. According to (7), it is concluded that

$$\begin{aligned} \chi_1 &= \lambda(\theta - \theta_d) + m_p r [L \sin(\beta + \theta_p) - \sin(\beta + \theta_{pd})] \\ &\quad + m_p L^2(\theta_p - \theta_{pd}) = 0, \\ \chi_2 &= 0, \end{aligned} \quad (52)$$

$$\chi_3 = m_p r [L \sin(\beta + \theta_p) - L_d \sin(\beta + \theta_{pd})] = 0.$$

Similarly, since s_2 converges to zero, one derives

$$\begin{aligned} s_2 &= x_{l_2}^{q_2/p_2} + a_2 x_{l_2} + c_2 x_{l_1} = 0 \\ \Rightarrow x_{l_1} &= m_p(L - L_d) = 0 \Rightarrow L = L_d. \end{aligned} \quad (53)$$

Substituting $L = L_d$ into χ_3 of (52), one has

$$m_p r [L_d \sin(\beta + \theta_p) - L_d \sin(\beta + \theta_{pd})] \Rightarrow \theta_p = \theta_{pd}. \quad (54)$$

Combining the results $L = L_d$, $\theta_p = \theta_{pd}$ and make some arrangements, one can derive from $\chi_1 = 0$ that

$$\lambda(\theta - \theta_d) = 0 \Rightarrow \theta = \theta_d. \quad (55)$$

Therefore, one can finally conclude from (53)–(55) that all the system state variables converge to the desired values. $\square$

Remark 2. As is well known, sliding mode control is efficient in dealing with uncertainties/disturbances. On the other hand, it also usually has chattering issue, which poses challenge to practical application. To address this problem, one first designs sliding mode observers to estimate and further compensate the disturbances. Consequently, the controller only needs to deal with the residual disturbances, whose amplitudes are much smaller than those of the disturbances. Furthermore, instead of directly designing the control input, one elaborately introduces a virtual input v . Although $\dot{v}$ contains *signed terms*, its integration, namely v , is still continuous. In this way, the real control inputs, which include v and other continuous terms, are also continuous.

5. Hardware experiment verification

In this section, hardware experiments are performed to verify the performance of the proposed method.

Fig. 2 illustrates the WBR platform in the laboratory. The upper leg length and the lower leg length are respectively designed as 250 mm and 246 mm. Two knee motors with the rated torque of 13 N m are utilized to support the floating body. The wheel radius is 50 mm, weighing about 1.5 kg. Two wheel hub motors have the rated torque of 2 N m. An X-sense IMU is mounted on the floating body to measure the incline angle



Fig. 2. WBR platform in the laboratory.

of the robot. C++ is utilized as the programming language to write the code of controller. The control system runs under the environment of Ubuntu 16.0.4. The sensors information is collected by an NVIDIA Jetson NX through CAN communication. The control period is set as 10 ms, which guarantees good real-time performance.

In order to comprehensively evaluate the performance of the proposed method, two sets of experiments were conducted. Specifically, the proposed method is first compared with existing methods. After that, different kinds of uncertainties and disturbances are exerted on the system, so as to test the robustness of the proposed method.

5.1. Comparison with existing methods

The proposed method is first compared with the SMC method (Zhou et al., 2021) and a model predictive control (MPC) method. It is important to note that, selecting control parameters for nonlinear controllers is a rigorous and meticulous process, necessitating extensive experience and numerous trials and errors. To guarantee fair comparison, the gains for each controller have been carefully tuned in the tests. Finally, the control parameters for the proposed method are elaborately tuned as

$$\rho_1 = 27, \rho_2 = 27, \rho_3 = 9, a_1 = 3, c_1 = 15, a_2 = 2, c_2 = 6,$$

$$k_{s_1} = 5, k_{a_1} = 10, k_{\rho_1} = 12, k_{s_2} = 3, k_{a_2} = 8, k_{\rho_2} = 10,$$

$$p_1 = 9, q_1 = 11, p_2 = 9, q_2 = 11.$$

The MPC method is designed based on linearized model in consideration of the computational efficiency. After plenty of tuning, the weight matrices of MPC are elaborately determined as:

$$Q = \text{diag}\{300, 100, 200, 200, 600, 800\}, R = \text{diag}\{1, 0.5\}.$$

The prediction horizon and control horizon of the MPC controller are set as

$$H_p = 40, H_c = 5.$$

Besides, the following system input constraint is considered:

$$|u(k_i)| \leq u_{\max},$$

where $u_{\max}$ denotes the upper bound of the input. To calculate the optimal control signals, the cost function is defined as

$$J(x_0) = \min_{i=0}^{N-1} q(x_i, v_i) + p(x_N), \quad (56)$$

wherein $J(x_0)$ and $p(x_N) = x_N^T Q_N x_N$ represent the cost function and terminal cost, respectively. Q_i is the solution of the discrete-time algebraic Riccati equation. $q(x_i, v_i) = (x_i - y_f^T(i))^T Q_i (x_i - y_f^T(i)) + v_i^T R v_i$ denotes the stage cost at the stage i . As for the SMC method (Zhou et al., 2021), its control gains are chosen as follows:

$$K_p = \text{diag}\{5.8, 4.8\},$$

$$K_d = \text{diag}\{0.6, 0.5\},$$

$$k = \text{diag}\{8.0, 4.2, 0.75, 1.1\}.$$

Table 1
Performance indices for the first group of experiments.

Indices	Our method	SMC method	MPC method
e_θ (rad)	1.2	7.9	3.0
e_L (m)	0.01	0.017	0.016
$\theta_{p\max}$ (rad)	0.07	0.13	0.12
θ_{pres} (rad)	0.01	0.07	0.05

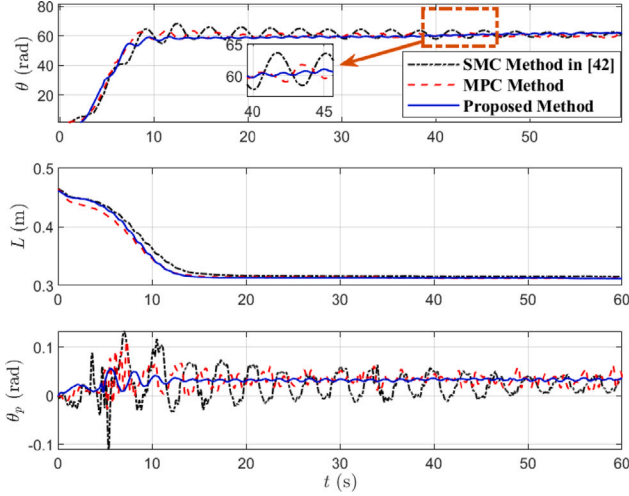


Fig. 3. Comparison with existing methods: (blue solid line) proposed method; (red dotted line) MPC method; (black dashdotted line) SMC method.

Besides, $\text{sat}(\cdot)$ is utilized to replace $\text{sgn}(\cdot)$ to alleviate chattering.

Some performance indices are defined as follows to better illustrate the control performance of the proposed method and the comparative methods:

- **Wheel Position Error** e_θ : robot wheel position error after the settling time t_s , i.e., $e_\theta = \max_{t > t_s} \{|\theta(t) - \theta_d|\}$.
- **Leg Length Position Error** e_L : robot leg length position error after the settling time t_s , i.e., $e_L = \max_{t > t_s} \{|L(t) - L_d|\}$.
- **Maximum robot incline angle** $\theta_{p\max}$: maximum robot incline angle throughout the control process, i.e., $\theta_{p\max} = \max_{t > 0} \{|\theta_p(t)|\}$.
- **Residual robot incline angle** θ_{pres} : maximum swing angle after the settling time t_s , i.e., $\theta_{pres} = \max_{t > t_s} \{|\theta_p(t)|\}$.

In the first group of experiments, one elaborately sets the slope angle β as 13 degrees instead of 0 (corresponding to flat ground), which can better test the balance performance of the above methods. The corresponding performance indices are recorded in Table 1. As is observed from Fig. 3, the proposed method can successfully stabilize the WBR even on the slope and considering leg length variation. The positioning errors of the wheel and leg length are respectively 1.2 rad and 0.01 m for the proposed method, which are evidently smaller compared with those of the SMC and MPC methods. Besides, the proposed method achieves much better balance performance than the comparative methods. Specifically, the maximum incline angle for the proposed method is about 0.07 rad (4 degrees). As for the SMC method and the MPC method, the maximum incline angles are respectively about 0.13 rad (7.4 degrees) and 0.12 rad (6.8 degrees). Furthermore, when the robot reaches the desired location, it almost stays steady with the help of the proposed method. On the contrast, the comparative methods still need to move back and forth to keep the robot balance, and obvious incline angle presents in this process (see Table 1 for details). Hence, it can be concluded that, the proposed method can stabilize the WBR more efficiently, which is very important when considering complex terrains.

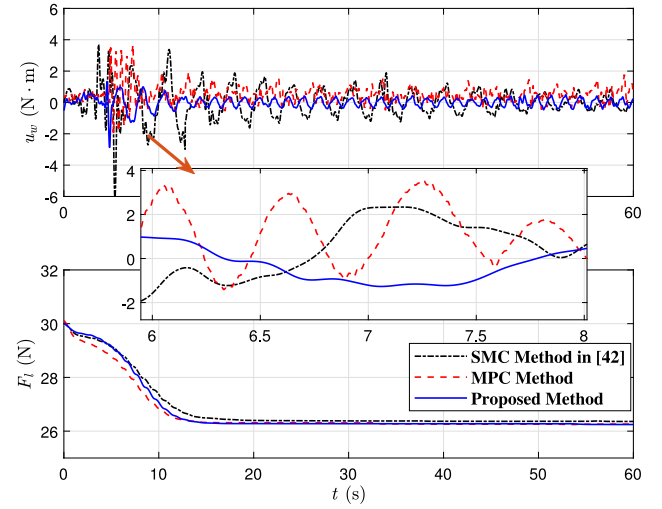


Fig. 4. Control inputs: (blue solid line) proposed method; (red dotted line) MPC method; (black dashdotted line) SMC method.

Fig. 4 depicts the curves of the control inputs. As a matter of fact, the WBR system is an inherently unstable system. Besides, there are inevitable uncertainties/disturbances, signal measuring noises to interfere the system. To eliminate such disturbances, the robot tends to move back and forth constantly to keep the robot balance, which leads to reasonable changes in the signals. The curves of control inputs may appear to change quickly. This is because one compresses 60 s of data in a small area. As is observed from Fig. 4, when one zooms out the subfigure of τ , and sees the details between 6–8 s, the curves of all the three methods are actually smooth. Therefore, one can conclude that the chattering problem can be successfully alleviated by the proposed algorithm. Given the similar nature of the control inputs and their non-focal role in this paper, they are omitted from subsequent experiments for brevity.

5.2. Robustness verification

The following 4 cases of experiments are performed to test the robustness of the proposed method:

Case 1: An extra payload (3 kg) is put on the robot, while the nominal system parameters and control gains are kept unchanged. Besides, nonzero initial incline angle is introduced to interfere the WBR. Finally, one intentionally strikes the robot two times respectively at about 50 s and 75 s (simulating sudden external disturbances).

Case 2: The robot keeps its balance while the leg length continuously changes (sinusoidal motion). Furthermore, two sudden strikes are exerted on the robot respectively at about 45 s and 70 s.

Case 3: The robot moves on a complex terrain. Specifically, it first moves on flat ground, and then climbs a slope of 17 degrees. On top of the slope, the robots meets a small platform. After that, it moves down a slope of -17 degrees, and finally returns to the flat ground. The considered terrain is shown in Fig. 2 behind the robot.

Case 4: The proposed method is applied to the 3-dimensional motion of the robot, instead of the planar motion. Specifically, the robot moves along a randomly-generated 2-dimensional curve on the ground, meanwhile the robot height is adjusted in real time.

The experiment results of the robustness verification are presented in Figs. 5–8. It can be observed from Fig. 5 that, the proposed method is barely influenced by the payload mass uncertainty and initial incline angle of the robot. Besides, the proposed method can re-stabilize the WBR quickly from the external disturbances even under such circumstances. Specifically, the incline angle of the robot is sufficiently suppressed within 6 degrees during the whole control process, and the

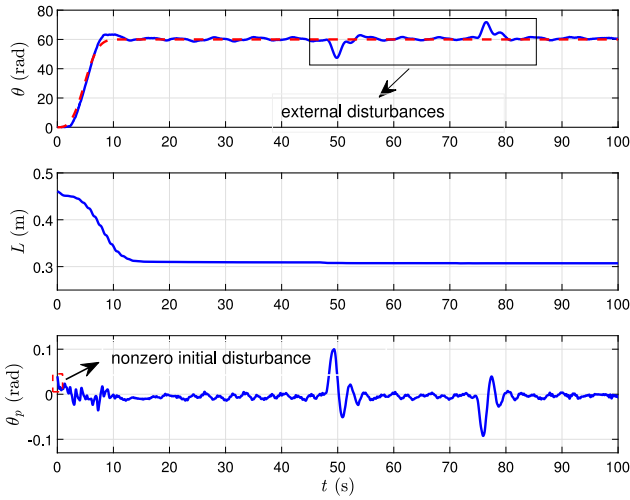


Fig. 5. Experiment results of Case 1.

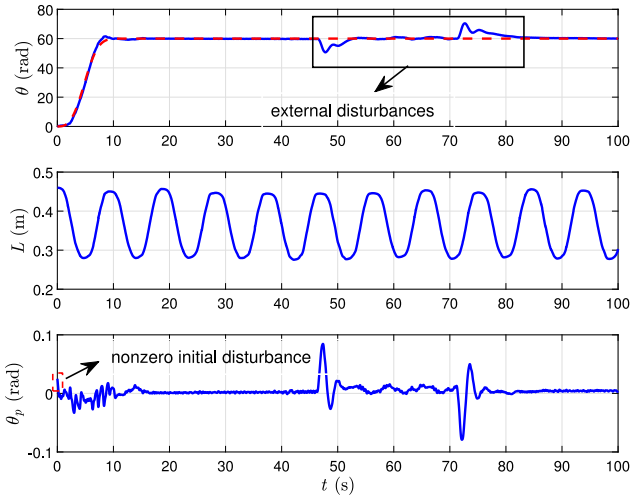


Fig. 6. Experiment results of Case 2.

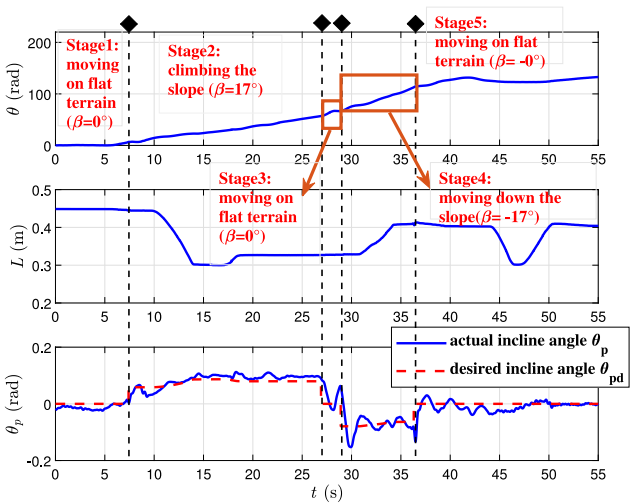


Fig. 7. Experiment results of Case 3.

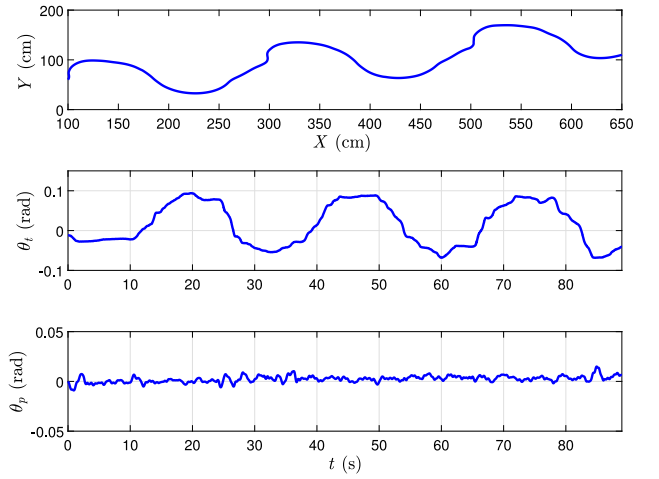


Fig. 8. Experiment results of Case 4.

the leg length is continuously changing. The two sudden strikes also do not cause much influence to the WBR, which shows good robustness of the proposed method. In Fig. 7, the WBR traverses a constantly varying slope while simultaneously adjusting its leg length. It is worth noting that, the reference trajectories of wheel and leg length are manually given through a remoter in real time, which are intentionally set to be not very smooth. Nevertheless, the balance of the robot is always well guaranteed. Particularly, the robot incline angle θ_p is always nearby θ_{pd} ². Such results demonstrate that the proposed method can adapt to different complex terrains. Finally, Fig. 8 presents a scenario where the WBR moves along a randomly generated curve on the ground while dynamically adjusting its height. The top subfigure shows the specified curve on the $X - Y$ plane (i.e., the ground). It is also noted that, there is a slight difference between the two leg lengths when the robot turns around. This is because one need a roll angle θ_t of the robot to provide centripetal force (see the second subfigure of Fig. 8), so that possible tumble can be avoided. From this point, one can say that the introduction of legs endows the WBR better maneuverability compared with traditional wheeled robots. One can see from the last subfigure of Fig. 8 that, the robot not only manages to perform fluent 3-dimensional motion (2-dimensional motion on the ground and 1-dimensional motion in the robot height), but also retains satisfactory balance performance (the incline angle θ_p is always suppressed within 0.03 rad throughout the control process).

In summary, the experimental results convincingly demonstrate the robustness, adaptability, and efficiency of the proposed method in maintaining the stability and balance of the WBR under various operating conditions and external disturbances. This is particularly important for the WBR, which is inherently unstable and often operates in complex terrains/environments.

6. Conclusion

With high movement efficiency and satisfactory adaptivity to uneven terrain, wheeled bipedal robot (WBR) has gained significant interest from researchers worldwide. Nevertheless, the inherent underactuation of the WBR system poses a tremendous challenge to achieving stable balance control. To tackle this challenge, a novel sliding mode control strategy is proposed in this paper. Specifically, through elaborate state transformations, the system dynamics are first restructured into a cascaded form. On this basis, sliding mode observers are then constructed to estimate the disturbances. Besides, sliding mode

adverse effects caused by disturbances are eliminated in about 6 s. Fig. 6 shows that, the robot can keep its balance very well even when

² The desired incline angle θ_{pd} is closely related to slope angle β .

controllers are designed to drive the system states to the sliding surfaces within fixed time. Through rigorous Lyapunov-based analysis, the desired equilibrium point is proved to be asymptotically stable. Thanks to the disturbance observer and novel design of sliding surfaces, the chattering problem of SMC is greatly alleviated, resulting in more smooth control actions and improved control performance. Hardware experiments have demonstrated that, the proposed method could be successfully applied to the 3-dimensional motion of WBR. Additionally, the proposed approach exhibits remarkable robustness when confronted with parameter uncertainties and external disturbances. In the future, endeavors will delve into the realms of jumping and pacing motions for WBRs, pushing the boundaries of their dynamic capabilities. Besides, more attempts will be made on the intelligent control of WBR.

CRedit authorship contribution statement

Biao Lu: Writing – review & editing, Writing – original draft, Validation, Methodology. **Haixin Cao:** Validation, Investigation. **Yongchun Fang:** Supervision. **Jing Zhang:** Validation. **Yunsong Hao:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.conengprac.2024.106108>.

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